Cluster Analysis of CART-Synthesised Data: A Factorial Simulation of k-Means and Ward's Linkage

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Abstract

Synthetic data allow information to be shared while protecting confidential records, but their usefulness depends on whether analyses of the synthetic data reproduce those of the original data. We investigate this for cluster analysis through a factorial simulation of 144 scenarios crossing separation, number of groups, dimensionality, correlation and distribution family (720,000 evaluations), applying k-means and Ward's linkage to data synthesised by sequential CART. Agreement between the numbers of clusters estimated from synthetic and original data rises with separation. Overall, k-means recovers the original number more often than Ward's method (0.84 vs. 0.81), on skewed Gamma data (0.81 vs. 0.73) but not on Normal data (0.87 vs. 0.89). A bootstrap null shows that this reflects the greater stability of the k-means estimate, not smaller damage from CART, and that resampling alone reproduces most centroid and dispersion distortion. A real-data case study agrees. Utility should be judged against a resampling reference.

Keywords: Synthetic data, CART synthesis, `synthpop`, k-means, Ward's linkage, Cluster stability, Bootstrap

1 Introduction

Synthetic data are artificial records produced by a generative model fitted to a confidential dataset, released in place of the original records so that the statistical structure of the data is retained while no real individual is exposed. The idea originates in statistical disclosure control, where Rubin (1993) proposed treating the confidential values as missing and releasing multiply-imputed replacements, and Little (1993) developed the companion analysis of masked data; Drechsler (2011) gives a book-length treatment of the resulting framework, and Matthews and Harel (2011) survey the wider family of disclosure-limitation methods within which synthesis sits. Synthetic data were initially developed mainly for statistical agencies, but their application has since expanded to many other fields. General-purpose synthesisers are now distributed as statistical software (Nowok, Raab, and Dibben, 2016; Templ, Meindl, Kowarik, and Dupriez, 2017), and administrative-data research is an active area of application (Kokosi, De Stavola, Mitra, Frayling, Doherty, Dove, Sonnenberg, and Harron, 2022). Synthetic records are also generated for health services research (Gonzales, Guruswamy, and Smith, 2023; Giuffrè and Shung, 2023), for biobehavioural science, where they additionally serve reproducibility and hypothesis generation (Quintana, 2020), and, increasingly, through deep generative architectures applied to tabular data (Goyal and Mahmoud, 2024; Zhao, Kunar, Birke, Van der Scheer, and Chen, 2024).

Synthetic data are mainly generated to support subsequent analyses, and cluster analysis is one of the most widely used unsupervised approaches: given unlabelled observations, it partitions them into groups that are internally homogeneous and mutually distinct. Two main approaches have dominated applied practice for decades: centroid-based methods, such as k-means (MacQueen, 1967; Steinley, 2006), which minimise the within-cluster sum of squares and remain the most widely deployed clustering algorithms (Ahmed, Seraj, and Islam, 2020); and agglomerative hierarchical methods, such as Ward's linkage (Ward, 1963), which build a dendrogram by successive merging and do not require the number of groups to be fixed in advance. Both remain central to contemporary applications: in single-cell transcriptomics, where partitions define cell types (Zappia and Oshlack, 2018), and in mass cytometry, where the choice among clustering methods materially changes the biological conclusion (Liu, Song, Wong, Zhang, Yu, Lin, and Ding, 2019).

These two areas intersect whenever a clustering analysis must be performed on data that cannot be released in raw form. In clinical and administrative settings, for example, subgroup discovery can be highly valuable, but access to the original data is often restricted for privacy reasons (Gonçalves, Ray, Soper, Stevens, Coyle, and Sales, 2020; El Emam, Mosquera, and Bass, 2020; Bhanot, Qi, Erickson, Guyon, and Bennett, 2021). In model development, access to the original data can also be slow or limited, and synthetic records then serve as an augmentation and development resource

(Rankin, Black, Bond, Wallace, Mulvenna, and Epelde, 2020). Synthetic data can therefore provide a practical solution in both situations. However, when an analysis is performed on synthetic rather than original data, it is important to assess whether the resulting partition is similar to the one that would have been obtained from the original records. This is what the *analytical utility* of synthetic data refers to.

Analytical utility can be assessed from a general and from a specific perspective. *General* utility asks whether the synthetic data resemble the original in a global sense, and *specific* utility asks whether a particular analysis yields matching conclusions (Snoke, Raab, Nowok, Dibben, and Slavkovic, 2018). The distinction matters because the two can diverge: a generator that reproduces marginal distributions and low-order correlations faithfully may still distort the higher-order geometry on which a specific procedure depends. Empirical work bears this out. Benchmarking studies of synthetic health records find that utility rankings depend heavily on which downstream task is used to measure them (Yan, Yan, Wan, Zhang, Omberg, Guinney, Mooney, and Malin, 2022; El Emam, Mosquera, Fang, and El-Hussuna, 2022), and direct replications of published analyses on synthetic counterparts show agreement that varies by study and by estimand (Reiner Benaim, Almog, Gorelik, Hochberg et al., 2020). A recent systematic review of tabular synthesis tools makes the same point from the measurement side, finding no consensus on which utility metrics should be reported and little agreement between them (Lautrup, Hyrup, Zimek, and Schneider-Kamp, 2024). Utility is therefore not a scalar property of a dataset; it is a property of the pairing between a dataset and the analysis applied to it.

Clustering provides a useful approach for assessing this relationship. Unlike a regression coefficient, which summarises a conditional mean and is comparatively robust to local perturbation, a partition is sensitive to the fine structure of the point cloud: relative positions, local density, separability and within-group dispersion. Two datasets can share identical means, variances and pairwise correlations and still yield entirely different partitions. Distortion that a generator introduces into local geometry may therefore appear in a clustering analysis before it appears in a resemblance check.

This article asks whether CART-based sequential synthesis, the default non-parametric engine of `synthpop` (Nowok et al., 2016) and a widely used method for tabular data, preserves the structure that clustering relies on. This study extends the line of inquiry opened by Chen (2025), which established a simulation harness for this question but restricted attention to k-means on Normal data and relied on a resemblance statistic that proved insensitive to structural quality. We broaden the scope in three directions: a second clustering method, Ward's linkage, which shares the within-cluster sum-of-squares criterion of k-means but optimises it by greedy agglomeration; a skewed distribution family that exposes the axis-aligned behaviour of tree-based synthesis; and a set of structural metrics that compare the partition recovered from the synthetic data with the partition recovered from the original data.

Whether a synthetic dataset leads the analyst to the same number of clusters as the original is a question about the stability of cluster analysis under perturbation, a topic with an extensive resampling literature (Ben-Hur, Elisseeff, and Guyon, 2002; Dudoit and Fridlyand, 2002; Tibshirani and Walther, 2005; Hennig, 2007; von Luxburg, 2010). We use that literature to calibrate our agreement measure: the reference point is not

perfect agreement but the agreement obtained when the synthetic data are replaced by a bootstrap resample of the original.

The utility of synthetic data for clustering analysis is assessed through three main objectives. First, to determine whether CART-synthesised data preserve the structure needed to recover the number of clusters, and how this depends on separation, number of groups, dimensionality, feature correlation and distribution family. Second, to determine whether the cluster structure is preserved beyond the number of clusters: how much fidelity of centroid location, within-group dispersion and group sizes is lost under synthesis, and which design factors drive that loss. Third, to determine whether partitioning and hierarchical methods respond similarly to synthetic distortion, or whether one is systematically more robust than the other. We address these questions with a factorial simulation. For each scenario, several original datasets are simulated from a mixture with known group structure, and each original dataset is then synthesised many times with CART. We compare the number of clusters recovered from each synthetic dataset with two references: the true number of groups used to simulate the original dataset, and the number of clusters recovered from that original dataset by the same algorithm. Each synthetic result is also set against the same quantity computed on bootstrap resamples of the original dataset, which separates the effect of CART from ordinary resampling variability. Finally, we apply the same comparison to a real dataset, the diabetes data of Reaven and Miller (1979).

2 Methods

The study is a controlled factorial simulation. We first define the data and the discrepancy being studied and the two clustering algorithms, then the generation of original and synthetic data, then the metrics used to quantify structural degradation and their bootstrap reference, and finally the way those metrics are summarised across replications.

2.1 Data and discrepancy

Each original dataset is an $N \times p$ data matrix $\mathbf{X}_R = (\mathbf{x}_1, \ldots, \mathbf{x}_N)^\top$ of N observations on p continuous variables, with rows $\mathbf{x}_i \in \mathbb{R}^p$ and columns (variables) $X_1, \ldots, X_p$, together with a vector of planted group labels $\mathbf{g} = (g_1, \ldots, g_N)^\top$, $g_i \in \{1, \ldots, k\}$, which is known by construction but is never used by the clustering algorithms. A generative mechanism G produces a synthetic $N \times p$ matrix $\mathbf{X}_S = G(\mathbf{X}_R)$ with the same variables. Let A be the downstream algorithm, producing summaries $\theta_R = A(\mathbf{X}_R)$ and $\theta_S = A(\mathbf{X}_S)$. We study the discrepancy

$$\Delta(\theta_R, \theta_S) = f(N, p, \rho, \sigma, k, F), \tag{1}$$

as a function of the sample size N, dimensionality p, feature correlation ρ, separation σ, number of latent groups k and distribution family F. A generator is useful for a task over a region of the design space when Δ remains small throughout it, and the content of an evaluation is the shape of Δ: which factors inflate it, which leave it unchanged, and whether different algorithms respond similarly.

In the clustering setting, θ comprises the estimated number of groups $\hat{k}$ together with the geometry of the recovered partition: its centroids, its group sizes and its within-group dispersion. Accordingly Δ is instantiated as a family of structural-fidelity measures defined in Section 2.4.

To study how synthesis affects cluster structure, we use simulations with known ground truth rather than real administrative data. Each scenario fixes a configuration $(N, p, k, \rho, \sigma, F)$ for which the true group membership of every point is known by construction. This makes it possible to separate the difficulty of the clustering problem itself from the distortion introduced by synthesis, avoids relying solely on resemblance-based evaluations, and allows Δ in Eq. (1) to be traced as an explicit function of each design factor.

2.2 Clustering algorithms

We compare two standard approaches: k-means (MacQueen, 1967) as a partitioning method, and agglomerative clustering with Ward's linkage (Ward, 1963) as a hierarchical method. k-means minimises the within-cluster sum of squares and so tends to favour convex, isotropic groups of comparable size (Steinley, 2006; Xiong, Wu, and Chen, 2009), with decision boundaries given by the bisectors of the centroid Voronoi tessellation; we retain the lowest-inertia solution over ten initialisations. Ward's linkage merges, at each step, the pair of clusters whose union least increases the total within-cluster error sum of squares. The two methods therefore share a variance-based criterion and differ in how they optimise it: Ward's merges are greedy and never revisited, so a merge driven by a local density irregularity early in the hierarchy propagates to the final partition, whereas k-means reassigns points iteratively. Because synthesis perturbs exactly such local irregularities, the two optimisation strategies may respond differently to it. Classical comparisons found that the sensitivity of clustering methods to error perturbation differs between methods and between types of perturbation, and that the behaviour of k-means depends on its starting configuration (Milligan, 1980). Both objectives are stated formally in Section S7 of the supplementary material.

2.3 Data generation

2.3.1 Original data

Each original matrix $\mathbf{X}_R$ contains $N = 1000$ observations drawn from a k-component mixture in $\mathbb{R}^p$ with known group labels and equal component sizes (N/k observations each, up to rounding). The minimum pairwise centroid distance is controlled by the separation σ: the first centroid $\boldsymbol{\mu}_1$ sits at the origin and each subsequent centroid $\boldsymbol{\mu}_j$ is drawn uniformly on the sphere of radius σ about the origin and accepted only if it lies at distance at least σ from all centroids already placed. The centroids are drawn afresh for every original dataset. Within-component covariance equals the $p \times p$ correlation matrix $\mathbf{R}$ with unit diagonal and off-diagonal entries ρ, giving uncorrelated (spherical) components at $\rho = 0$ and correlated (elliptical) components at $\rho = 0.4$; separation is therefore expressed in units of the within-component standard deviation. In the *Normal* family, components are multivariate Gaussian with mean $\boldsymbol{\mu}_j$ and covariance $\mathbf{R}$. In the *Gamma* family, independent Gamma variates (shape = 1.5, rate = 5) are

Table 1 Factors of the simulation design and the question each addresses. Crossing all five gives 144 scenarios; with $n = 5$ independent original datasets per scenario and $m = 1000$ synthetic datasets generated from each, the design yields 720,000 evaluations.

Factor	Levels	Rationale
Separation σ	0.1, 2, 6, 10	Spans overlapping to clearly distinct groups; tests how much separation between groups is needed before synthesis can preserve the structure.
Number of groups k	2, 3, 4	More groups increase the complexity of the cluster structure that the generator needs to preserve.
Dimensionality p	2, 5, 10	Each synthesised dimension compounds the axis-aligned leaf structure of the generator.
Correlation ρ	0, 0.4	Tests whether dependence between features degrades a variable-by-variable synthesis.
Distribution family F	Normal, Gamma	Contrasts symmetric Gaussian density with skewed Gamma density.
Sample size N	1000 (fixed)	Held constant so that variation is attributable to the factors above.

standardised to zero mean and unit variance within each component, correlated using the Cholesky factor of $\mathbf{R}$ (Johnson, 1987, chap. 4), and shifted to the component centroid. The two families thus share their first two moments within components and differ only in shape: before correlation is induced, each Gamma margin has skewness $2/\sqrt{1.5} \approx 1.63$, producing right-skewed groups that stress the axis-aligned nature of CART.

The factors we vary are the same high-level geometric properties used to parameterise cluster-analysis benchmarks generally, namely overlap between groups, their number, the ambient dimension and the shape of the components (Qiu and Joe, 2006; Zellinger and Bühlmann, 2025); here they serve not to benchmark clustering algorithms against each other but to locate the regions in which synthesis preserves the structure those algorithms depend on.

Table 1 lists the design factors. Each factor is varied to examine how it affects the conditions under which synthesis succeeds or fails: crossing all five gives $3 \times 4 \times 3 \times 2 \times 2 = 144$ scenarios. For each scenario, $n = 5$ independent original datasets are simulated, and each original dataset is synthesised $m = 1000$ times, resulting in $144 \times 5 \times 1000 = 720{,}000$ synthetic-versus-original evaluations.

Expected directions.

The rationales in Table 1 imply the following expected directions, which the analysis examines. Both agreement measures should increase with separation σ. Recovery of the planted number of groups should decrease with k. Geometric distortion should increase with p, because each synthesised coordinate adds its own leaf structure. Correlated components ($\rho = 0.4$) should be harder for a variable-by-variable synthesiser than

uncorrelated ones, and skewed Gamma components harder than Normal ones. For the comparison between the two algorithms no direction was fixed in advance: they share a criterion and differ in how they optimise it (Section 2.2).

2.3.2 Synthetic data (CART)

For every original dataset, synthetic data $\mathbf{X}_S = G(\mathbf{X}_R)$ are generated with the CART engine of `synthpop` (Nowok et al., 2016), using proper synthesis (the trees are fitted to a bootstrap sample of the original data) and a minimum leaf size of ten. Variables are synthesised sequentially in the order $X_1, \ldots, X_p$. The first variable is sampled from its observed marginal distribution. Each subsequent variable X_j is then modelled by a regression tree (Breiman, Friedman, Olshen, and Stone, 1984, chap. 8) on the already-synthesised variables, and synthetic values are drawn from the empirical distribution of the observed values in the corresponding leaf,

$$X_j^{\text{syn}} \sim \widehat{P}\big(X_j \mid \text{leaf}(X_1^{\text{syn}}, \ldots, X_{j-1}^{\text{syn}})\big). \tag{2}$$

This makes the method flexible and distribution-free, but it imposes an axis-aligned, piecewise-constant structure on the joint distribution: the synthetic point cloud is assembled from rectangular blocks whose boundaries are the tree splits, and synthetic values are resampled observed values. When the original data are smooth and symmetric, these blocks tile the space finely enough to be barely visible; when the data are skewed or heavy-tailed, they can become visible as a hardening of smooth density gradients into discrete steps. Detecting this failure mode, and measuring how strongly it propagates into the recovered partition, is the purpose of the evaluation. The group label is appended as the last variable of the synthesis sequence, so it does not influence the synthesis of the features; it is never used by the clustering algorithms.

Software and seeds.

Original data were generated and synthesised under R 4.3 (R Core Team, 2024) with `synthpop` 1.9.2 (Nowok et al., 2016), `mvtnorm` 1.4.2 and `arrow` 25.0.0; the `rpart` version of that run was not recorded, and the synthetic draws were reproduced bit for bit with `rpart` 4.1.24 under R 4.3.3. Clustering and evaluation used Python 3.12 with `scikit-learn` 1.9.0 (Pedregosa et al., 2011), `numpy` 2.5.1, `scipy` 1.18.0, `pandas` 3.0.5 and `pyarrow` 25.0.0. All seeds derive from a base seed of 16. The 144 scenarios are shuffled after `set.seed(16)`, one seed per scenario is then drawn after a second `set.seed(16)`, and the five replicates of a scenario are generated in turn from that seed. The synthetic draw m of replicate r uses the seed $16 + 10^7 g + 10^5 r + m$, where g is the position of the scenario in the list of original file names sorted by R; because R's `sort()` is locale-dependent, this index, and hence the synthetic data, is reproduced only under the `en_US.UTF-8` collation (the C collation gives a different index), and the distributed per-scenario seed table records g explicitly. Every k-means fit, on original and synthetic data alike, uses `random_state = 42` with ten initialisations, so the original and synthetic fits share their starting points. Refitting the synthetic data with independent seeds (replicate 1 and the first 20 draws of every scenario, 2,880 pairs) left $\hat{k}_S$ unchanged in 95.7% of pairs and changed AG by less than 0.001 (0.832

versus 0.831) and MCD, MVD and ΔG by at most 0.002, so the shared seed does not drive the results. Both stages are embarrassingly parallel over scenarios and were distributed across 18 worker processes on a single multi-core workstation (Section S8 of the supplementary material).

2.4 Evaluation metrics

Structural utility is quantified by two count-agreement measures and three geometric-fidelity measures, each set against a bootstrap reference (Section 2.5). All variables are standardised before clustering. The scaler is fit on the original data and applied unchanged to the corresponding synthetic dataset, so that all geometric quantities are expressed in a common coordinate system; standardising each dataset separately would rescale away part of the very distortion the metrics are meant to detect (see Section 5.2).

Number of groups.

For a dataset and an algorithm, the number of groups is estimated by maximising the average silhouette width (Rousseeuw, 1987) over the fixed candidate range $\{2, \ldots, 8\}$, applied identically to original and synthetic data. Earlier work (Chen, 2025) used the range $\{2, \ldots, \min(k+3, 8)\}$, which ties the search to the planted k; the count analyses under that range are reported as a sensitivity analysis in Section S3 of the supplementary material. We write $\hat{k}_R$ for the estimate obtained from an original dataset and $\hat{k}_{S,m}$ for the estimate obtained from the m-th synthetic dataset generated from it, and $\hat{k}_{R,m}$ for the estimate from the original dataset that generated synthetic dataset m. For a configuration with M evaluations we report two agreement proportions,

$$\mathrm{RR} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\big[\hat{k}_{S,m} = k\big], \qquad \mathrm{AG} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\big[\hat{k}_{S,m} = \hat{k}_{R,m}\big], \tag{3}$$

where $\mathbf{1}[\cdot]$ is the indicator function. The recovery rate RR compares the synthetic estimate with the planted number of groups. It therefore confounds two sources of failure: distortion introduced by synthesis, and the inability of the clustering procedure to recover the planted structure even from the original data (for instance, when the groups overlap so strongly that a single diffuse cloud is a reasonable description). The agreement AG compares the synthetic estimate with the estimate from the original data, obtained with the same algorithm and the same selection rule, and so removes the first source. It does not remove a third: $\hat{k}$ varies from sample to sample even without synthesis, so part of any disagreement is ordinary sampling instability of the silhouette estimate. AG is therefore read against the bootstrap null AG_0 of Section 2.5. RR, together with the recovery rate of the original data itself, $\mathbf{1}[\hat{k}_R = k]$, is reported to separate the sources of failure.

Geometric fidelity.

The geometric metrics are computed with the number of clusters fixed at the planted k on both sides, so that they describe the geometry of the partition conditional on

the correct count. They require the groups recovered from $\mathbf{X}_S$ to be matched to those from $\mathbf{X}_R$, since cluster labels are arbitrary. We build the matrix of pairwise centroid distances between the k original and k synthetic clusters and solve the resulting linear assignment problem (Kuhn, 1955), obtaining the one-to-one matching that minimises total centroid distance. All fidelity metrics are computed on the aligned clusters.

Group-balance error (ΔG).

This metric evaluates whether the distribution of observations across clusters is preserved under synthesis. It is defined as

$$\Delta G = \left| G_S - G_R \right|, \tag{4}$$

where G_R and G_S are the Gini coefficients of the cluster-size vectors of the partitions recovered from the original and synthetic data. For sizes $\{n_1, \ldots, n_k\}$,

$$G = \frac{\sum_{a=1}^{k} \sum_{b=1}^{k} |n_a - n_b|}{2k \sum_{a=1}^{k} n_a}, \tag{5}$$

with $G = 0$ for perfectly equal groups. A value of ΔG close to 0 indicates that synthesis has preserved the distribution of cluster sizes. Because the planted groups have equal sizes, G_R reflects only the imbalance that the algorithm itself produces on the original data.

Mean centroid distance (MCD).

This metric measures the average distance between the centroids of the original clusters and the centroids of the corresponding synthetic clusters. With aligned original and synthetic centroids $\boldsymbol{\mu}_j^R$ and $\boldsymbol{\mu}_j^S$,

$$\mathrm{MCD} = \frac{1}{k} \sum_{j=1}^{k} \left\| \boldsymbol{\mu}_j^R - \boldsymbol{\mu}_j^S \right\|, \tag{6}$$

so small values indicate that synthesis preserves the location of the groups.

Mean variance difference (MVD).

This metric evaluates whether the synthetic clusters retain a dispersion similar to that of the original clusters. Writing v_j^R and v_j^S for the mean squared distance of points to their centroid in aligned original and synthetic clusters,

$$\mathrm{MVD} = \frac{1}{k} \sum_{j=1}^{k} \left| v_j^R - v_j^S \right|. \tag{7}$$

Scale-free versions.

Both MCD and MVD are expressed in the units of the scaler, which is fit on the pooled original data; because the spread of the pooled data grows with the separation between

groups, a standardised unit corresponds to a larger within-component distance at large σ. We therefore also report two scale-free versions. The relative variance difference

$$\mathrm{RVD} = \frac{1}{k}\sum_{j=1}^{k}\left|\frac{v_j^S}{v_j^R} - 1\right| \tag{8}$$

is the average relative error of the within-cluster dispersion, and $\mathrm{MCD}_{\mathrm{sd}} = \mathrm{MCD}/s_w$ expresses the centroid displacement in units of the within-component standard deviation $s_w = \{\sum_i \|\mathbf{x}_i - \boldsymbol{\mu}_{g_i}\|^2/(Np)\}^{1/2}$ of the original dataset in the scaled coordinates, where $\boldsymbol{\mu}_{g_i}$ is the mean of the planted group of observation i. We also report MVD/p, since v_j is a sum over the p coordinates.

2.5 Bootstrap null

Proper synthesis in `synthpop` first draws a bootstrap sample of the original data and then fits the CART models to it. A bootstrap resample of the original data is therefore the synthesis procedure without the tree step, and it is the natural reference for how much disagreement ordinary resampling alone produces. For each of the 720 original datasets we draw $B = 30$ bootstrap resamples of its N rows, apply the scaler fitted on the original data unchanged, and run the same silhouette scan, giving estimates $\hat{k}_b$, $b = 1, \ldots, B$. The bootstrap agreement

$$\mathrm{AG}_0 = \frac{1}{B}\sum_{b=1}^{B}\mathbf{1}\left[\hat{k}_b = \hat{k}_R\right] \tag{9}$$

is averaged over datasets in the same way as AG, and each method is compared with its own null through the excess $\mathrm{AG} - \mathrm{AG}_0$: a negative excess means that synthesis loses agreement beyond what resampling already loses. The same resamples give a *geometric floor* for every fidelity metric: MCD, MVD, ΔG, RVD and $\mathrm{MCD}_{\mathrm{sd}}$ computed, at the planted k, between the partition of the original data and the partition of the bootstrap resample, averaged over the B resamples. The excess of an observed metric over its floor is the part of the distortion attributable to CART beyond resampling. Resample b of original dataset r in scenario s is drawn with the `numpy` generator seeded by $(20260929, s, r, b)$, and the same resample is used for both algorithms, so that the comparison between methods is paired. With $B = 30$, the Monte Carlo standard error of a scenario-level AG_0 is at most about 0.04.

2.6 Statistical analysis

The design is hierarchical: $m = 1000$ synthetic datasets are nested within each of $n = 5$ original datasets, which are nested within each of the 144 scenarios. Agreement proportions are reported directly over evaluations. Fidelity metrics are aggregated first to a scenario mean and then averaged across scenarios; the bands and error bars in the figures are the mean ± 1.96 standard errors of the scenario-level means. They describe the spread of the metric across the design cells rather than sampling uncertainty, and

using the scenario as the unit avoids treating the m synthetic draws from one original dataset as independent. Algorithm comparisons, and comparisons of each method with its bootstrap null, use Wilcoxon signed-rank tests across the 144 scenario-level means; scenarios with equal means are dropped as ties. To rank the design factors, we fit a main-effects linear model to the 720 replicate-level agreement proportions (one per original dataset) and report partial η^2 for each factor.

For visualisation we use the two-dimensional scenarios, plotting the standardised features against each other; no dimensionality reduction is applied, so the panels show the geometry the algorithms actually operate on. In the paired original/synthetic displays, cluster labels are matched by the same Hungarian alignment used for the fidelity metrics and then oriented so that colour tracks vertical position consistently across panels.

3 Results

3.1 The block-like artefact

Figures 1 and 2 show a two-dimensional, two-group scenario at moderate separation ($\sigma = 2$): the first original replicate of the scenario in the top row and its first CART synthetic draw below, each partitioned by both algorithms. Reading across a row shows how the two algorithms divide the same cloud; reading down a column shows what synthesis did to that division.

On Normal data (Fig. 1) the overall cluster structure is largely preserved after synthesis. Because CART resamples observed values within leaves, the synthetic cloud contains repeated coordinates, visible as short vertical and horizontal runs of points, most clearly near the edges of the range (panel d). Both algorithms divide the synthetic cloud along a boundary of similar orientation to the one they find in the original, although the Ward partition of the synthetic data (panel f) assigns a visibly smaller share of points to the upper group. At $\sigma = 2$ the true groups overlap substantially. Neither algorithm reproduces the true labelling exactly: where the true groups interpenetrate (panel a), both algorithms replace them with a sharp linear boundary through the overlap region (panels b and c).

On Gamma data (Fig. 2) the block-like artefact introduced by CART becomes more visible. The original groups are right-skewed, with a dense corner and long tails running out along both axes. In the synthetic cloud those tails break into discrete, axis-aligned patches, and the dense corner is reproduced as a set of rectangular blocks rather than a continuous gradient. This is the characteristic artefact of sequential tree synthesis: the piecewise-constant leaf structure of Eq. (2) can become more apparent for skewed, non-spherical densities. The partitions still divide the cloud in roughly the same orientation, but the boundary is displaced and the resulting groups have different internal spread, the distortion that the fidelity metrics quantify below.

3.2 Number of groups

The number of clusters estimated from synthetic data can be evaluated against two references, each addressing a different question (Section 2.4). Comparison with the

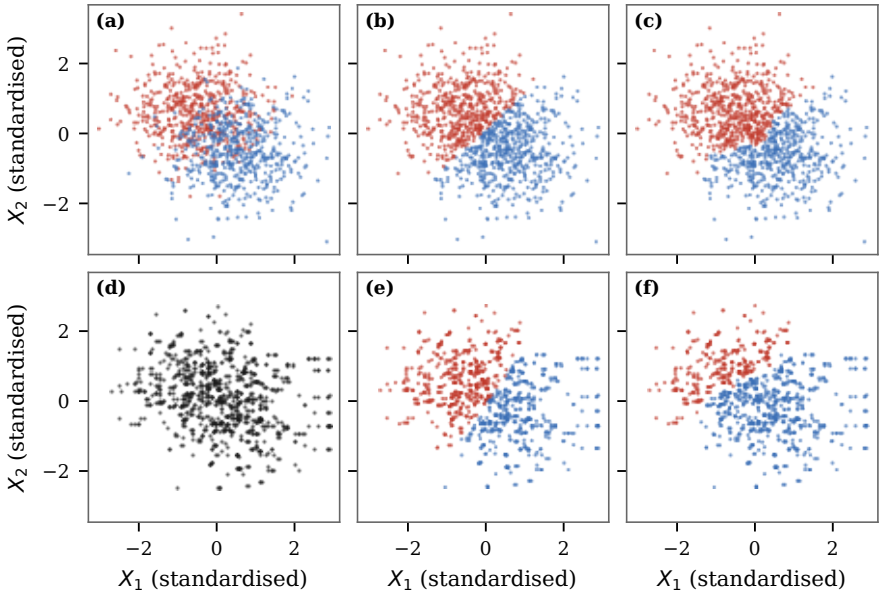


Fig. 1 Normal (spherical) components: geometry largely preserved under synthesis. Top row, original data: (a) true group labels, (b) k-means partition, (c) Ward partition. Bottom row, the corresponding CART synthetic dataset: (d) synthetic group label, (e) k-means partition, (f) Ward partition. The data shown are replicate 1 of the scenario $k = 2$, $p = 2$, $\rho = 0$, $\sigma = 2$ and its first synthetic draw; axes are the two standardised features. In the partition panels, colours are matched across rows by Hungarian alignment, so the same colour denotes corresponding groups in the original and synthetic panels. Panel (d) is drawn in a single colour: the group label in the synthetic data is itself synthesised, so it is a prediction from the synthetic features rather than a planted truth, and it has no bijective correspondence with the original groups. Both rows contain the same number of observations ($N = 1000$); the synthetic panels can look sparser only because CART resamples observed values, so that points coincide more often.

planted number of groups (RR) assesses whether the synthetic data support recovery of the underlying truth, and so combines synthesis distortion with the inability of the clustering procedure to find the planted structure at all. Comparison with the number estimated from the original data (AG) removes the second component but retains the ordinary resampling instability of $\hat{k}$, which the bootstrap null measures (Section 3.3). Both references are reported throughout, and Figure 3 shows each against separation, stratified by the true number of groups.

Separation is the decisive factor under both references. Against the planted number of groups, recovery rises from 0.247 (k-means) and 0.281 (Ward) at $\sigma = 0.1$, through 0.288 and 0.295 at $\sigma = 2$, to 0.810 and 0.789 at $\sigma = 6$ and 0.954 and 0.950 at $\sigma = 10$. The transition is sharp: up to $\sigma = 2$ recovery is low and dominated by the overlap of the groups themselves, while at $\sigma = 10$ it is high for every k.

Against the number estimated from the original data the picture is markedly more favourable, and the gap is largest exactly where the groups overlap: agreement is 0.684 (k-means) and 0.646 (Ward) at $\sigma = 0.1$, rising to 0.982 and 0.975 at $\sigma = 10$, for

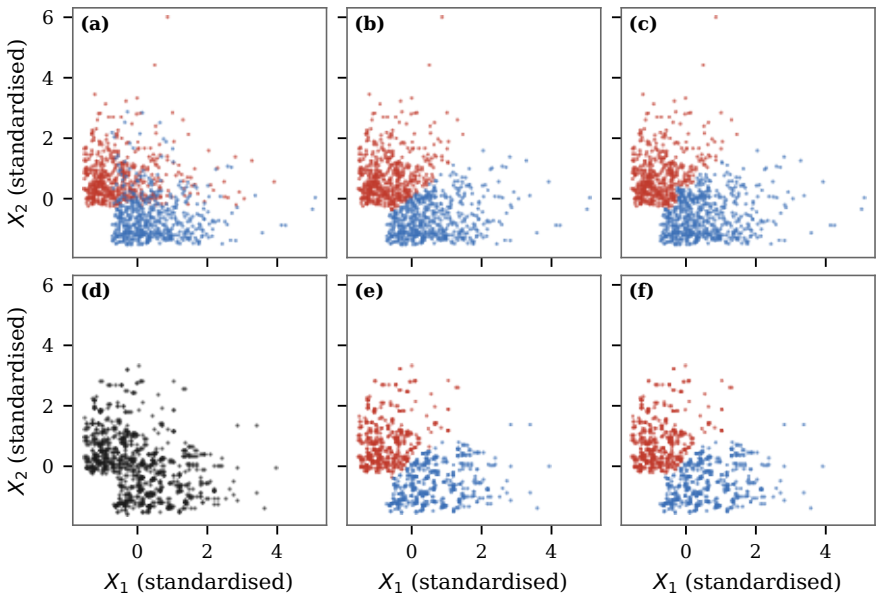


Fig. 2 Gamma (right-skewed) components: the same synthesis mechanism breaks smooth tails into axis-aligned blocks. Panel layout (a)–(f), colour matching, the single-colour panel (d), the replicate and draw shown, and the scenario ($k = 2$, $p = 2$, $\rho = 0$, $\sigma = 2$) are as in Fig. 1, and both rows again contain $N = 1000$ observations. Comparing panel (d) with panel (a) shows the artefact directly: the continuous density gradient of the original becomes a set of discrete rectangular patches.

overall agreement of 0.839 and 0.811. The explanation is that at low separation the algorithms frequently fail to recover the planted count from the original data as well: at $\sigma = 0.1$, k-means finds the true k in only 0.233 of original datasets and Ward in 0.294. Much of what the first reference records as a failure of synthesis is therefore a failure of clustering under overlap, which synthesis reproduces.

The number of groups behaves very differently under the two references. Against the planted count, recovery is much higher for fewer groups: 0.819 for k-means at $k = 2$, against 0.468 at $k = 3$ and 0.437 at $k = 4$; in the overlapping regime the three- and four-group cases are almost never recovered (at most 0.12), yet they exceed 0.90 once $\sigma = 10$. Against the original data, this pattern largely disappears (0.847, 0.838 and 0.833). The difficulty of larger k thus lies mainly in the clustering problem, which synthesis reproduces, rather than in synthesis itself. The partial η^2 values from the main-effects model make the contrast explicit: for the planted reference, separation (0.565 for k-means, 0.577 for Ward) and k (0.287 and 0.357) dominate, whereas for agreement with the original data separation (0.245 and 0.229) is followed by correlation (0.141 and 0.075) and distribution family (0.019 and 0.105), with k below 0.001. For agreement, however, the main effects explain only a third of the variation ($R^2 = 0.340$ and 0.349); adding all two-way interactions raises R^2 to 0.606 and 0.605, with large correlation-by-separation terms (0.282 and 0.179), so the ranking describes average

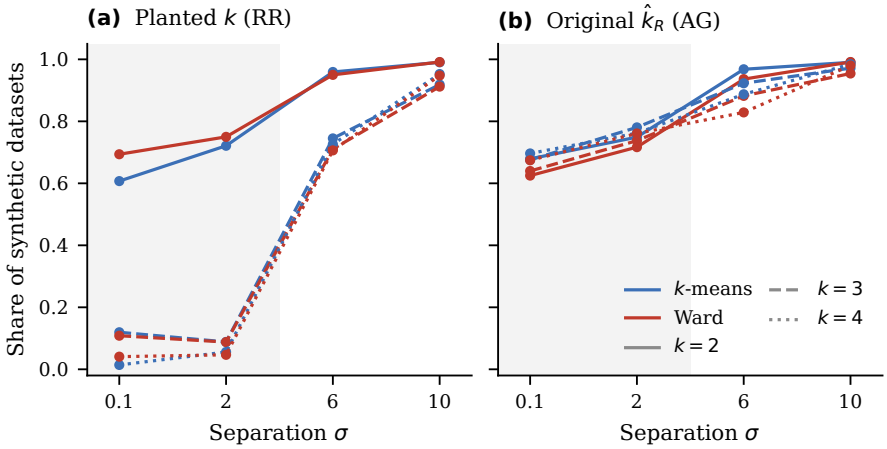


Fig. 3 Cluster-count agreement between synthetic and original data, against separation and stratified by the true number of groups, under two reference conventions: (a) agreement with the number of groups planted by the simulation (RR); (b) agreement with the number the same algorithm recovers from the original data (AG). The separation levels are shown as categories because they are not equally spaced; the shaded region marks overlapping groups ($\sigma \leq 2$). In (a), agreement is low while the groups overlap for $k \geq 3$ and moderate for $k = 2$; (b) stays higher throughout and depends far less on k, because in the overlapping regime the algorithms often fail to recover the planted count from the original data as well. Candidate range $\{2, \ldots, 8\}$; each point pools 60,000 original–synthetic pairs.

tendencies rather than a separable structure. Table 2 reports the marginal means for every design factor under both references.

Three further patterns in Table 2 deserve comment. First, agreement with the original data is higher for correlated components (0.925 versus 0.753 for k-means; 0.877 versus 0.744 for Ward), but this is an effect of the estimator rather than of synthesis. At $\sigma \leq 2$ the silhouette scan returns $\hat{k}_R = 2$ for 91.1% (k-means) and 87.8% (Ward) of original datasets with $\rho = 0.4$, against 38.3% and 62.2% with $\rho = 0$, so original and synthetic estimates coincide largely because both collapse onto the smallest candidate. Where the estimator is not collapsed ($\sigma \geq 6$), agreement is slightly lower for correlated components (0.948 versus 0.960 for k-means; 0.914 versus 0.944 for Ward). Second, dimensionality has opposite effects in the two regimes. At $\sigma \leq 2$ Ward's agreement rises with p (0.536, 0.703 and 0.839 at $p = 2$, 5 and 10), again because of collapse: at $p = 10$ and $\sigma \leq 2$ the scan selects $\hat{k}_R = 2$ for 80.8% (k-means) and 90.0% (Ward) of original datasets. At $\sigma = 6$, where the groups are separated but not trivially so, agreement falls with p (k-means 0.976, 0.903 and 0.899; Ward 0.976, 0.853 and 0.818). Third, the collapse cells marked in Table 2, a sixth of the design, contribute near-automatic agreement and raise the overall values. The distribution family produces a consistent gap under both references, which Section 3.5 examines.

3.3 Agreement relative to the bootstrap null

Table 3 sets AG beside its bootstrap null. Both algorithms agree with the original estimate less often after synthesis than after bootstrap resampling: overall AG_0 is

Table 2 Agreement in the number of groups by design factor, over the full 720,000-evaluation design, under the two references of Eq. (3): the planted number of groups (RR) and the number estimated from the original data (AG). Each entry is a marginal mean over all other factors; candidate range $\{2, \ldots, 8\}$. Wilcoxon p: paired signed-rank test of k-means against Ward across the 144 scenario-level means. The capped range $\{2, \ldots, \min(k+3, 8)\}$ is reported in Supplementary Table S2.

		Planted k (RR)		Original $\hat{k}_R$ (AG)	
Factor	Level	k-means	Ward	k-means	Ward
Separation σ	0.1[a]	0.247	0.281	0.684	0.646
	2[a]	0.288	0.295	0.764	0.738
	6	0.810	0.789	0.926	0.882
	10	0.954	0.950	0.982	0.975
Number of groups k	2	0.819	0.846	0.847	0.817
	3	0.468	0.455	0.838	0.803
	4	0.437	0.435	0.833	0.811
Dimensionality p	2	0.605	0.613	0.817	0.756
	5	0.554	0.563	0.849	0.806
	10	0.565	0.560	0.851	0.870
Correlation ρ	0	0.553	0.567	0.753	0.744
	0.4	0.597	0.590	0.925	0.877
Distribution F	Normal	0.593	0.604	0.869	0.890
	Gamma	0.557	0.554	0.809	0.731
Overall		0.575	0.579	0.839	0.811
Wilcoxon p		0.750		5.4×10^{-5}	

[a] All collapse cells lie at $\sigma \leq 2$: scenarios with $k \geq 3$ in which every original dataset gives $\hat{k}_R = 2$ (22 of 96 for k-means, 26 of 96 for Ward). They contain 15.3% and 18.1% of all evaluations, and agreement is 0.955 and 0.929 inside them against 0.799 and 0.762 outside (Supplementary Section S6).

0.871 for k-means and 0.826 for Ward, so the excess is -0.032 and -0.016. Synthetic agreement lies below the null in 99 of the 144 scenarios for k-means (10 above, 35 ties; Wilcoxon $p = 1.1 \times 10^{-11}$) and in 90 for Ward (30 above, 24 ties; $p = 4.8 \times 10^{-6}$). The loss attributable to CART beyond resampling is therefore real but small, a few points of agreement, against a resampling instability that is already sizeable: even a bootstrap resample of the original fails to reproduce $\hat{k}_R$ in 12.9% of cases for k-means and 17.4% for Ward. Two independent original datasets from the same scenario, a coarser reference that also changes the centroid geometry, agree in 0.801 and 0.767 of pairs (Supplementary Section S4).

3.4 Geometric fidelity

Figure 4 traces the geometric-fidelity metrics against separation, together with their bootstrap floors, and Table 4 summarises them. Mean centroid distance falls steeply

Table 3 Agreement with the original estimate after CART synthesis (AG) and after bootstrap resampling of the original data (AG_0, $B = 30$ resamples per original dataset), and the excess $AG - AG_0$; negative values mean that synthesis loses agreement beyond resampling. Candidate range $\{2, \ldots, 8\}$. The excess is computed before rounding.

		k-means			Ward		
Stratum	Level	AG	AG_0	Excess	AG	AG_0	Excess
Separation σ	0.1	0.684	0.721	−0.038	0.646	0.644	+0.002
	2	0.764	0.814	−0.051	0.738	0.754	−0.016
	6	0.926	0.960	−0.034	0.882	0.923	−0.040
	10	0.982	0.989	−0.007	0.975	0.984	−0.009
Distribution F	Normal	0.869	0.880	−0.011	0.890	0.902	−0.012
	Gamma	0.809	0.863	−0.053	0.731	0.750	−0.020
Normal, σ	0.1	0.665	0.655	+0.011	0.782	0.773	+0.009
	2	0.879	0.903	−0.024	0.873	0.886	−0.013
	6	0.947	0.974	−0.028	0.924	0.964	−0.040
	10	0.983	0.988	−0.005	0.982	0.986	−0.004
Gamma, σ	0.1	0.702	0.788	−0.086	0.511	0.515	−0.004
	2	0.648	0.726	−0.078	0.603	0.623	−0.019
	6	0.906	0.946	−0.040	0.841	0.881	−0.041
	10	0.981	0.991	−0.010	0.969	0.983	−0.014
Overall		0.839	0.871	−0.032	0.811	0.826	−0.016

as groups separate, from 0.604 (k-means) and 0.859 (Ward) at $\sigma = 0.1$ to 0.090 and 0.095 at $\sigma = 10$, and mean variance difference falls from 0.353 and 0.613 to 0.112 and 0.133. Observed values are lower for k-means in 133 of the 144 scenarios for MCD, in all 144 for MVD and in 138 for ΔG (paired Wilcoxon $p < 10^{-20}$ for each).

Most of this distortion, however, is already produced by resampling. Bootstrap resampling alone reproduces 0.222 of the 0.303 observed MCD for k-means and 0.403 of 0.460 for Ward, 0.150 of 0.219 and 0.324 of 0.392 for MVD, and 0.021 of 0.030 and 0.053 of 0.061 for ΔG, and the floor already places k-means below Ward (in 121, 135 and 132 of the 144 scenarios for MCD, MVD and ΔG). CART adds a positive excess in almost every scenario (MCD 144 and 143 of 144 for k-means and Ward; MVD 144 and 141; ΔG 144 and 129; $p < 10^{-20}$ for each), but the excess is not smaller for k-means: it is larger for MCD (0.081 versus 0.057; Wilcoxon $p = 8.8 \times 10^{-6}$) and equal for MVD (0.069 versus 0.068; $p = 0.973$) and nearly equal for ΔG (0.009 versus 0.008; $p = 0.462$, but $p = 0.002$ at $n = 50$). The lower geometric distortion of k-means on synthetic data is therefore inherited from its lower distortion under resampling, not produced by a smaller effect of CART.

The scale-free metrics correct the reading of the separation trend. In units of the within-component standard deviation, the centroid displacement still falls with separation (k-means 0.605 to 0.189; Ward 0.860 to 0.201), and CART moves the centroids by about 0.10 (k-means) and 0.08 (Ward) within-component standard deviations beyond resampling. Dispersion behaves differently. The relative variance difference shows that

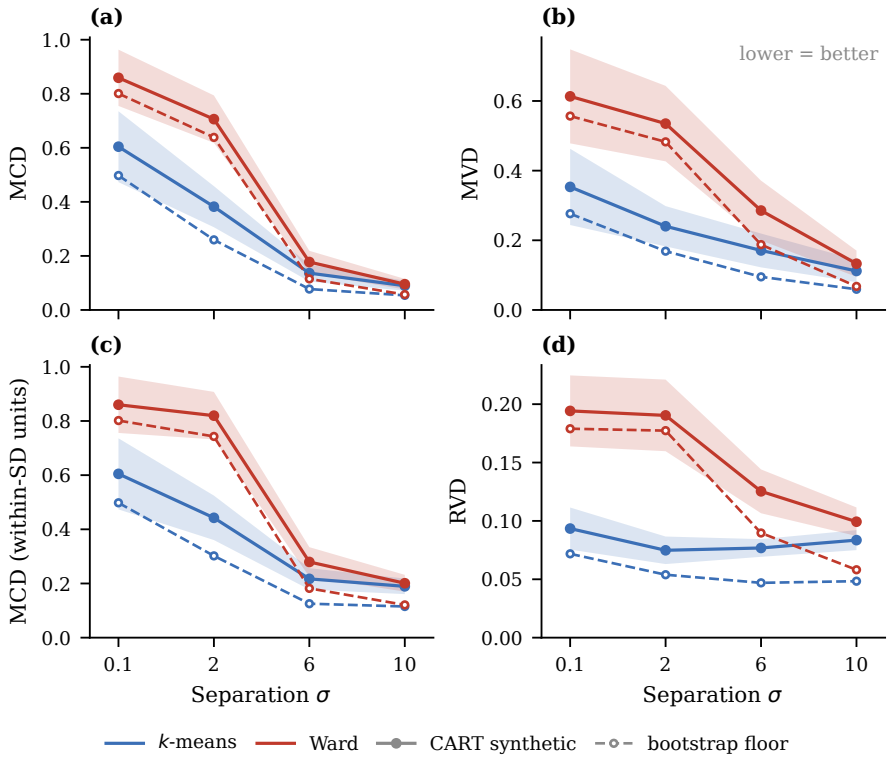


Fig. 4 Geometric fidelity against separation, observed after CART synthesis (solid lines, filled markers) and under bootstrap resampling of the original data (dashed lines, open markers; $B = 30$). (a) Mean centroid distance and (b) mean variance difference, in the units of the scaler fit on the original data; (c) mean centroid distance in units of the within-component standard deviation (MCD_{sd}); (d) relative variance difference (RVD). Points are means of the 36 scenario-level means at each separation level; bands are ± 1.96 standard errors across those scenarios and describe the spread over the design, not sampling uncertainty. The vertical distance between a solid and a dashed line of the same colour is the excess attributable to CART.

a synthetic cluster's dispersion deviates from the original's by 8.2% on average for k-means and 15.2% for Ward, of which 5.5 and 12.6 points are resampling; the excess attributable to CART is about 2.7 points for both (0.027 versus 0.026; $p = 0.891$). This excess does not decline with separation; it grows (k-means 0.022, 0.021, 0.030 and 0.035 across the four levels of σ; Ward 0.015, 0.013, 0.036 and 0.041). The decline of absolute MVD with σ is thus mostly a decline of the floor and a change of units, not a reduction of the dispersion distortion that CART introduces.

Total dispersion distortion grows with dimensionality (Supplementary Fig. S3a): MVD rises from 0.06 at $p = 2$ to 0.39 at $p = 10$ for k-means and from 0.13 to 0.65 for Ward. Because v_j in Eq. (7) sums squared deviations over the p coordinates, MVD/p is roughly constant (k-means 0.032, 0.040 and 0.039 at $p = 2, 5, 10$; Ward 0.067, 0.078 and 0.065) and the floor grows in the same way, so much of the accumulation is resampling noise summed over coordinates; the per-coordinate excess attributable

Table 4 Geometric fidelity after CART synthesis (observed), under bootstrap resampling of the original data (floor, $B = 30$) and their difference (excess), averaged over the 144 scenario means, at the planted k. MCD, MVD and ΔG are in the units of the scaler fit on the original data; $\mathrm{MCD}_{\mathrm{sd}}$ is in within-component standard deviations and RVD is a relative error. Last column: Wilcoxon p for the excess of k-means against the excess of Ward across the 144 scenarios. Values by separation, distribution family and dimensionality are in Supplementary Section S5.

	k-means			Ward			Excess
Metric	Observed	Floor	Excess	Observed	Floor	Excess	p
MCD	0.303	0.222	0.081	0.460	0.403	0.057	8.8×10^{-6}
MVD	0.219	0.150	0.069	0.392	0.324	0.068	0.973
ΔG	0.030	0.021	0.009	0.061	0.053	0.008	0.462
$\mathrm{MCD}_{\mathrm{sd}}$	0.363	0.260	0.103	0.540	0.462	0.078	2.7×10^{-5}
RVD	0.082	0.055	0.027	0.152	0.126	0.026	0.891

to CART increases only mildly with p (k-means 0.008, 0.011 and 0.013; Ward 0.006, 0.010 and 0.014).

Group balance is better preserved on Normal than on Gamma data (Supplementary Fig. S3b; ΔG 0.028 versus 0.032 for k-means, 0.057 versus 0.065 for Ward). k-means yields lower ΔG than Ward (in 138 of 144 scenarios; 0.030 versus 0.061), but the floor shows the same ordering (0.021 versus 0.053): with equal planted group sizes, the cluster sizes of a k-means partition fluctuate less under resampling, and hence under synthesis, consistent with the tendency of the k-means criterion to produce clusters of similar size (Xiong et al., 2009). Whether k-means would understate a genuine size imbalance is left to future work (Section 5.3).

3.5 Distribution family

The distribution family is the factor that exposes the CART artefact most directly. Figure 5 stratifies count agreement by distribution and separation under both references, each shown with its reference on data without synthesis. Against the planted number of groups (panels a and b), the two families show similar recovery while the groups overlap and diverge at intermediate separation: at $\sigma = 6$, k-means recovers the planted count 86.2% of the time for Normal data but only 75.9% for Gamma data (Ward: 84.5% versus 73.2%). Almost all of this gap, however, is already present in the original data, where the planted count is recovered in 90.0% versus 80.0% of datasets (k-means; Ward 91.1% versus 78.9%). It therefore reflects mainly how difficult skewed groups are to cluster, not what synthesis does to them.

Against the original data (panels c and d), the Gamma penalty is concentrated where the groups overlap and is much larger for Ward. At $\sigma = 0.1$, Ward's agreement is 0.782 for Normal but only 0.511 for Gamma data, and at $\sigma = 2$ it is 0.873 versus 0.603; the corresponding values for k-means are 0.665 versus 0.702 and 0.879 versus 0.648. At $\sigma = 6$ the gap narrows (k-means 0.947 versus 0.906; Ward 0.924 versus 0.841), and at $\sigma = 10$ a small gap remains for Ward (0.982 versus 0.969; k-means 0.983

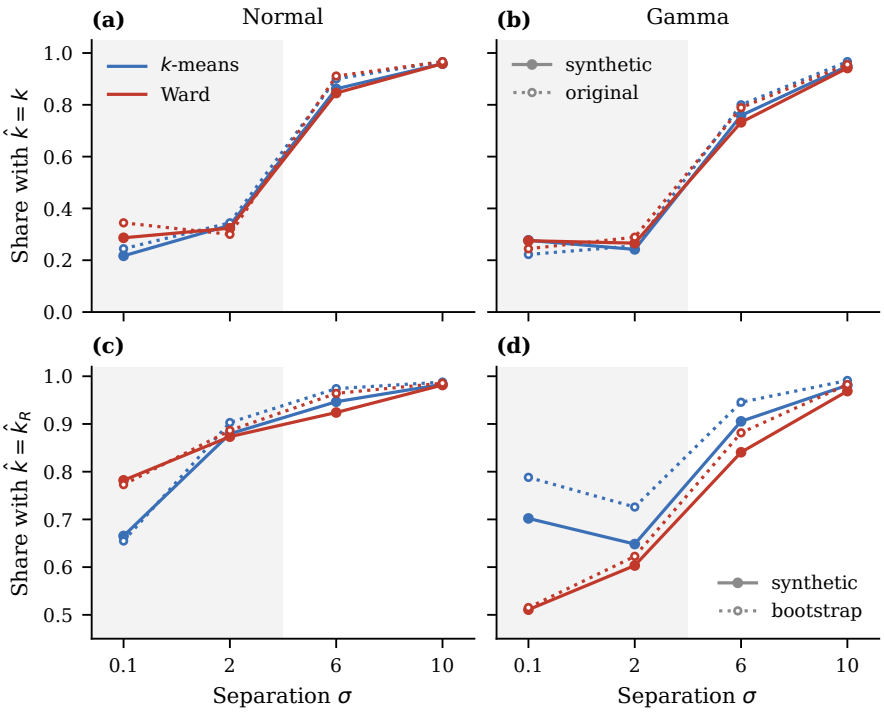


Fig. 5 Cluster-count agreement by distribution family (columns: Normal, Gamma) and separation, under the two references (candidate range $\{2, \ldots, 8\}$). (a, b) Share of synthetic datasets with $\hat{k}_S = k$ (RR; solid) beside the share of original datasets with $\hat{k}_R = k$ (dotted, open markers). (c, d) Share of synthetic datasets with $\hat{k}_S = \hat{k}_R$ (AG; solid) beside the share of bootstrap resamples with $\hat{k}_b = \hat{k}_R$ (AG_0; dotted, open markers). The vertical distance between a solid and a dotted line of the same colour is the part of the disagreement attributable to CART. Separation is shown as a categorical axis; the shaded region marks $\sigma \leq 2$.

versus 0.981). The bootstrap null shows where this penalty comes from. For Ward, the null already contains it: AG_0 is 0.515 for Gamma and 0.773 for Normal data at $\sigma = 0.1$, so the low agreement of Ward on skewed, overlapping data is an instability of its silhouette estimate that synthesis inherits, and the Gamma excess is small at every separation (between -0.041 and -0.004). For k-means the null is higher on Gamma data (0.788 at $\sigma = 0.1$ and 0.726 at $\sigma = 2$), and the loss attributable to CART is largest exactly there (-0.086 and -0.078).

This addresses the first objective. CART preserves the structure needed to recover the number of groups on Normal data to within about one point of the bootstrap null (excess -0.011 for k-means and -0.012 for Ward), and its hardening of skewed densities lowers agreement on Gamma data beyond resampling, most markedly for k-means when groups overlap. The larger Gamma penalty seen for Ward in the raw agreement is mainly the instability of its estimate on skewed, overlapping data rather than a larger effect of CART.

3.6 Agreement between the two algorithms

Whether the two methods differ in robustness to synthetic distortion was the third objective, and the answer depends on the reference. Against the planted number of groups the two are indistinguishable: overall recovery is 0.575 for k-means and 0.579 for Ward, a paired Wilcoxon signed-rank test across the 144 scenario-level means finds no significant difference ($W = 3508.5$, $p = 0.750$), and the two algorithms reach the same outcome in 92.0% of individual evaluations.

Against the original data, k-means reproduces the number of groups more often than Ward: overall agreement is 0.839 versus 0.811, k-means is higher in 90 of the 144 scenarios (30 lower, 24 ties), and the paired test is significant ($W = 2088.5$, $p = 5.4 \times 10^{-5}$). For an analyst who will cluster a synthetic release, this is the practically relevant comparison, and it is not uniform. The advantage of k-means arises on Gamma data (0.809 versus 0.731; higher in 59 of 72 scenarios, $p = 2.6 \times 10^{-6}$), where it is significant at $\sigma = 0.1$ (0.702 versus 0.511, $p = 0.002$), $\sigma = 6$ (0.906 versus 0.841, $p = 0.001$) and $\sigma = 10$ (0.981 versus 0.969, $p = 0.006$) but not at $\sigma = 2$ (0.648 versus 0.603, $p = 0.099$). On Normal data there is no difference (0.869 versus 0.890, $p = 0.595$), and in the most overlapping Normal scenarios Ward is numerically higher (0.782 versus 0.665 at $\sigma = 0.1$; $p = 0.136$). The conclusion is unchanged under the capped candidate range (Supplementary Section S3).

The bootstrap null shows why. The same advantage is already present without synthesis: AG_0 is 0.871 for k-means against 0.826 for Ward ($W = 966$, $p = 6.0 \times 10^{-5}$), and 0.863 against 0.750 on Gamma data ($p = 6.3 \times 10^{-7}$), while on Normal data the nulls do not differ (0.880 versus 0.902, $p = 0.900$). The silhouette-based estimate of k is more stable under resampling for k-means, particularly on skewed data, and synthesis inherits that stability. Beyond resampling, k-means loses slightly more agreement than Ward (difference in excess -0.017, $W = 2846$, $p = 0.040$), again only on Gamma data (-0.034, $p = 0.024$; concentrated at $\sigma = 0.1$, -0.082, and $\sigma = 2$, -0.058), with no difference on Normal data ($p = 0.640$). A replication with $n = 50$ original datasets per scenario gives the same difference with much stronger evidence (-0.018, $p = 1.2 \times 10^{-5}$; Supplementary Section S12). The higher agreement of k-means on synthetic data thus reflects the greater stability of its $\hat{k}$, not smaller damage from CART, and the same holds for the geometric metrics (Section 3.4).

As a complementary check on internal cohesion, the difference in mean silhouette width between synthetic and original partitions at the planted k is small and centred near zero at every separation (at most 0.0125 in absolute value, for Ward at $\sigma = 0.1$). The absolute levels are nearly identical (k-means 0.396 versus 0.397; Ward 0.376 versus 0.377), suggesting that the apparent clustering quality is similar before and after synthesis. This is reassuring, but it also means that silhouette alone would fail to flag the geometric distortions the other metrics detect, an instructive negative result for anyone tempted to validate a synthetic release with an internal index.

4 Case study: the Reaven–Miller diabetes data

To illustrate the conclusions from the simulation study, we apply the same procedure to the diabetes data (Reaven and Miller, 1979). The data comprise 145 patients classified clinically as normal (76), chemical diabetes (36) or overt diabetes (33). Following Reaven and Miller (1979), we use three of the five variables in the distributed data: the areas under the plasma glucose and insulin curves during an oral glucose tolerance test, and the steady-state plasma glucose (SSPG), taken from the copy of the data distributed with the R package `rrcov` (Todorov and Filzmoser, 2009). The three variables are right-skewed (skewness 1.8, 1.8 and 0.7), and glucose area and SSPG are strongly correlated ($r = 0.77$). As in the simulation, the data were standardised with the means and standard deviations of the original, and $m = 1000$ synthetic datasets were generated with the settings of the simulation: CART with proper synthesis and a minimum leaf size of ten, with the variables synthesised in the order glucose, insulin and SSPG and the clinical class last. The bootstrap null used $B = 1000$ resamples of the original rows. Uncertainty is Monte Carlo uncertainty over draws only (R 4.5.3, `synthpop` 1.9.3, `rpart` 4.1.27; evaluation with the code of the simulation). In the synthetic data the block artefact of Figs. 1–2 appears as runs of repeated values and empty bands (Fig. 6a–b).

Reaven and Miller (1979) identified three groups, a central body and two wings in a rotating PRIM-9 display, and a three-group nearest-mean clustering (Friedman and Rubin, 1967) reproduced the clinical classification closely (84, 35 and 26 patients, with 20 patients reassigned). There the number of groups was fixed in advance; estimated from the data, as in our design, the silhouette selects $\hat{k} = 2$ for both methods, but only by a small margin over three groups (average silhouette width 0.571 versus 0.547 for k-means, 0.567 versus 0.544 for Ward). The original data thus sit close to the boundary between two and three groups, and the selected $\hat{k}$ is expected to be unstable under any perturbation. This near tie governs the synthetic results (Fig. 6c): the synthetic data reproduce $\hat{k} = 2$ in 62.9% of draws for k-means and 63.2% for Ward (AG), and the two methods do not differ (exact McNemar test on the paired draws, 58 versus 61 discordant draws, $p = 0.85$). Resampling the original rows gives almost the same agreement ($\mathrm{AG}_0 = 0.667$ and 0.637), so the excess over the null is -0.038 for k-means and -0.005 for Ward (Monte Carlo standard error 0.021), neither distinguishable from zero.

The partition-level metrics use the clinical $k = 3$, the analogue of the planted k in the simulation. To measure whether the same patients are grouped together without matching synthetic to original records, each synthetic dataset is partitioned with $k = 3$, every original patient is assigned to the nearest of its cluster means (Dudoit and Fridlyand, 2002; Tibshirani and Walther, 2005), and the adjusted Rand index (ARI; Hubert and Arabie, 1985) compares this assignment with the original partition of the same 145 patients. ARI is 1 for identical partitions and near 0 for chance agreement; for Ward, whose partition is not a nearest-centroid one, the ceiling is 0.955, which the bootstrap null includes.

All metrics show the same qualitative pattern as the simulation, with mostly larger magnitudes (Table 5). Every geometric metric exceeds its bootstrap floor for both

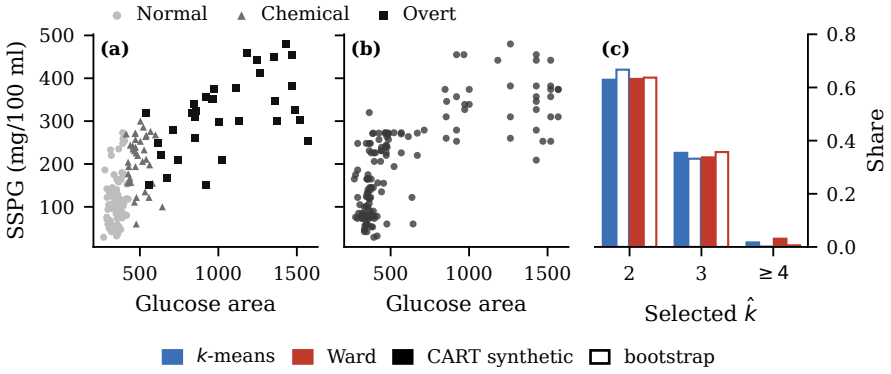


Fig. 6 Diabetes data, $N = 145$. (a) Original data by clinical class. (b) One CART synthetic dataset (single colour, as in Figs. 1–2). (c) Silhouette-selected $\hat{k}$ over $m = 1000$ synthetic datasets (filled) and $B = 1000$ bootstrap resamples (open); on the original data both methods select $\hat{k} = 2$.

methods by more than four Monte Carlo standard errors. ARI is 0.856 for k-means and 0.797 for Ward, below its bootstrap null by 0.066 and 0.040: synthesis also moves patients between groups beyond what resampling does, as it does over the whole simulation (0.036 and 0.011 below the null; Supplementary Section S10). Against the clinical classes, the original partitions give an ARI of 0.504 (k-means) and 0.536 (Ward), the partitions transferred from the synthetic data 0.481 and 0.471, and those from resamples 0.495 and 0.480: the synthetic data recover the diagnostic groups slightly less well than a resample. As the clinical classes were defined mainly from the glucose tolerance test, this comparison is partly circular.

As in the simulation, the count of groups is limited mainly by how stable $\hat{k}$ is on the original data, while CART adds a measurable distortion of the partition itself. The advantage of k-means in AG seen for skewed simulated data does not appear here. This is consistent with its explanation: k-means is only slightly more stable under resampling (AG_0 0.667 versus 0.637; exact McNemar test on the paired resamples, 86 versus 56 discordant, $p = 0.015$), and synthesis removes that margin (AG 0.629 versus 0.632). In all six orders of the three variables (Supplementary Section S11) every geometric excess is positive, every ARI excess negative and the AG of the two methods within 0.05; count agreement falls furthest below its null when glucose area is synthesised last (AG excess down to -0.12 for k-means and -0.14 for Ward).

5 Discussion

This study evaluated whether CART-generated synthetic data preserve the structural properties that cluster analysis relies on, comparing k-means and Ward's hierarchical clustering across a 144-scenario factorial design and 720,000 synthetic-versus-original evaluations, each read against a bootstrap resample of the same original data. Four findings emerge.

Table 5 Diabetes case study: means over $m = 1000$ synthetic datasets and $B = 1000$ bootstrap resamples, and the excess (synthetic minus bootstrap). AG uses $\hat{k}$, the other metrics $k = 3$; ARI (clinical) compares the transferred partition with the clinical classes. Monte Carlo standard errors of the excess: 0.021 for AG, 0.002–0.010 otherwise.

	k-means			Ward		
Metric	Synthetic	Bootstrap	Excess	Synthetic	Bootstrap	Excess
AG	0.629	0.667	−0.038	0.632	0.637	−0.005
ARI	0.856	0.922	−0.066	0.797	0.837	−0.040
MCD	0.383	0.229	0.154	0.472	0.348	0.124
MVD	0.216	0.165	0.051	0.313	0.242	0.070
ΔG	0.062	0.042	0.019	0.084	0.073	0.011
RVD	0.208	0.150	0.058	0.305	0.235	0.070
ARI (clinical)	0.481	0.495	−0.014	0.471	0.480	−0.009

First, separation dominates agreement in the number of groups. Agreement with the original data rises from 0.684 (k-means) and 0.646 (Ward) when the groups overlap to 0.982 and 0.975 when they are well separated, and recovery of the planted count rises from 0.247 and 0.281 to 0.954 and 0.950. The number of groups strongly affects recovery of the planted count (0.819 at $k = 2$ against 0.468 and 0.437 at $k = 3$ and 4, for k-means) but has little effect on agreement between original and synthetic estimates. Most of the effect of k is a property of the clustering problem that synthesis reproduces rather than a distortion that synthesis introduces. Where groups overlap, a sixth of the design consists of collapse cells in which the silhouette scan returns two groups for every original dataset, and agreement there is close to automatic.

Second, for a researcher who clusters a synthetic release, k-means reproduces the number of groups found in the original data more often than Ward's linkage (0.839 versus 0.811). This happens on skewed Gamma data (0.809 versus 0.731), at every separation except $\sigma = 2$, and not on Normal data (0.869 versus 0.890). The bootstrap null shows that this advantage reflects the greater stability of the k-means estimate of k under resampling (AG_0 0.871 versus 0.826), not smaller damage from CART: the agreement lost to synthesis beyond resampling is small for both methods (0.032 and 0.016) and, if anything, slightly larger for k-means. The geometry tells the same story. k-means shows lower centroid, dispersion and cluster-size distortion on synthetic data, but most of each metric is reproduced by resampling alone, and the excess attributable to CART is not smaller for k-means (larger for MCD, equal or nearly equal for MVD, ΔG and RVD). For an applied analyst, the practical implication is that the choice of algorithm matters more for the fidelity of a synthetic-data analysis than accuracy against a ground truth would suggest, and that the reason is the stability of the estimator rather than the generator.

Third, CART is not utility-neutral, and its effect depends on the underlying distribution. On Normal data the count agreement after synthesis is within about one point of the bootstrap null; on skewed Gamma data CART hardens smooth density gradients into axis-aligned blocks, visibly distorting geometry and lowering agreement

beyond resampling, most for k-means when groups overlap. CART adds a positive excess over the resampling floor to every geometric metric in almost every scenario; in relative terms it changes within-cluster dispersion by about 2.7% beyond resampling, and this relative excess grows rather than shrinks with separation.

Fourth, and methodologically, neither an internal validity index nor raw agreement is a sufficient check. Silhouette values remain almost unchanged even where centroid and dispersion fidelity degrade. A practitioner who validates a synthetic release by confirming that its silhouette profile matches the original may conclude that structure is intact in precisely the settings where this analysis shows it is not. Raw agreement, in turn, mixes the effect of the generator with the instability of the estimator, and the comparison between algorithms changes once each method is set against its own resampling null. The planted truth, the natural choice in a simulation, answers yet another question, and in this study it masked the difference between the two algorithms.

An interesting corollary of the design is that clustering can be read in the other direction, as an instrument rather than an object of study. Because a partition responds to local density, separability and dispersion simultaneously, the agreement between partitions recovered from original and synthetic data functions as a sensitive summary of geometric resemblance, one that reacts to distortions which marginal and bivariate checks may not detect. We did not pursue this here, but the metrics of Section 2.4 could be repurposed as general-utility diagnostics rather than task-specific ones.

5.1 Relation to previous work

The study extends Chen (2025) in three ways: a second algorithm that shares the criterion of k-means but optimises it differently, a skewed distribution family that exposes tree-synthesis artefacts, and structural metrics that compare partitions recovered from the original and synthetic data. It should be distinguished from the cluster-based utility measure of Woo, Reiter, Oganian, and Karr (2009), which clusters the merged original and masked records and measures how well each cluster mixes the two sources; that measure uses clustering as a device for general utility, whereas we ask whether a clustering analysis of the release reproduces the analysis of the original. The conclusion that CART is a reasonable default for well-behaved data but degrades on skewed distributions is consistent with the broader `synthpop` literature (Nowok et al., 2016; Snoke et al., 2018), and the qualitative block artefact matches the known axis-aligned bias of sequential tree synthesis. The finding that utility rankings depend on the downstream task echoes benchmarking work in the health-records setting (Yan et al., 2022; El Emam et al., 2022), where the same generator changes position depending on which analysis is used to score it, and an applied study of wearable physiological data in mental-health monitoring similarly found that the clustering utility of `synthpop` releases depends on the synthesis settings (Vidiella Villegas and López, 2025). The lower stability of Ward's estimate under resampling is consistent with, although not directly comparable to, the classical perturbation experiments of Milligan (1980), which found that methods differ in their sensitivity to error perturbation. A parallel question arises when values are missing rather than synthesised: Aschenbruck, Szepannek, and Wilhelm (2023) evaluate how imputation strategies affect the

clusters subsequently recovered from mixed-type data, and reach a structurally similar conclusion, that the imputation model and the clustering method interact and cannot be assessed in isolation. Since CART underlies both sequential synthesis and widely used non-parametric imputation, the two literatures are closer than their vocabularies suggest.

The count-agreement measure AG is, in effect, a stability criterion. A long line of work validates clusterings, and chooses their number, by asking how reproducibly a partition is recovered when the data are perturbed: by subsampling (Levine and Domany, 2001; Ben-Hur et al., 2002), by prediction from one half of the sample to the other (Dudoit and Fridlyand, 2002; Lange, Roth, Braun, and Buhmann, 2004; Tibshirani and Walther, 2005; Wang, 2010), or by the bootstrap (Jain and Moreau, 1987; Hennig, 2007; Fang and Wang, 2012); von Luxburg (2010) reviews these ideas. A synthetic release is a perturbation of the same kind, except that the synthesiser, not the analyst, produces it, so AG measures the stability of $\hat{k}$ under synthesis. Two results from this literature govern how AG should be read. First, stability is not correctness: Ben-David, von Luxburg, and Pál (2006) show that stability is driven by the structure of the clustering objective rather than by the true number of groups, so AG must be read together with recovery of the planted k; the collapse cells of Section 3.2, where agreement is high because the estimator returns two groups regardless, are an instance. Second, agreement under perturbation has no natural zero, because $\hat{k}$ is already variable on the original data. We therefore calibrate AG against a bootstrap null, in the spirit of resampling-based assessments of cluster structure (Suzuki and Shimodaira, 2006; Maitra, Melnykov, and Lahiri, 2012): agreement that a bootstrap resample of the original already fails to reach cannot be attributed to CART. In this study the calibration changes the comparison between algorithms, because the advantage of k-means in AG is matched by its advantage in AG_0. On the analysis side, the closest analogue is validation of a clustering on independent data (Ullmann, Hennig, and Boulesteix, 2022); a synthetic dataset can be regarded as a validation sample that the data holder is able to release.

5.2 Limitations

The evaluation rests on controlled simulations with known ground truth, complemented by a single small real dataset (Section 4), and its conclusions are scoped to the grid studied: correlations up to $\rho = 0.4$, a single Gamma shape, $N = 1000$ and exactly equal group sizes. This isolates mechanisms cleanly but limits direct generalisation to data with missingness, mixed types, unequal group sizes and unknown data-generating processes. The generator is limited to CART with a single parameterisation (minimum leaf size ten, fixed variable order) and the algorithms to k-means and Ward. Sample size was fixed at $N = 1000$ in the simulation, so small-sample behaviour is examined only through the case study ($N = 145$). The number of groups is selected by the silhouette criterion over $\hat{k} \geq 2$, so a single-cluster solution can never be chosen, even when the groups overlap so strongly that one cluster is the most reasonable description; other selection rules (Milligan and Cooper, 1985) may respond differently to synthesis. Count agreement is inflated where the estimator collapses onto the smallest candidate, as happens at low σ, high p and $\rho = 0.4$, and should be read

together with the bootstrap null and the geometric metrics. The bootstrap null rests on $B = 30$ resamples per original dataset, so comparisons of small excesses between the algorithms carry Monte Carlo error; a replication with ten times as many original datasets confirms them (Supplementary Section S12). Every k-means fit uses the same seed; a check with independent seeds (Section 2.3.2) showed no material effect. The Wilcoxon tests and the ± 1.96 standard-error bands are computed across the 144 scenarios of a designed grid, so they compare methods over that grid rather than over a population of scenarios.

Two conventions in the definition of MCD and MVD deserve emphasis, because both affect the numbers themselves and not only their interpretation. The first is what the metrics compare. Here they compare the partition recovered from the original data against the partition recovered from the synthetic data, which is the quantity a claim about synthesis requires; the superficially similar practice of comparing a recovered partition against the planted labels of the same dataset measures clustering accuracy instead, and carries no information about the generator. The second is scale. Both metrics are computed after standardising with a scaler fit on the pooled original data and applied unchanged to the synthetic data, so both are expressed in units of the pooled original feature scale. Standardising each dataset separately would rescale away part of the distortion the metrics are meant to capture, but the common scaler has its own dependence on the design: the pooled spread grows with σ, so a standardised unit is a larger within-component distance at large separation, and absolute MCD and MVD decline with σ partly for that reason. Neither convention is free of σ; only relative or within-component versions are. We therefore report RVD and $\mathrm{MCD}_{\mathrm{sd}}$ beside the absolute metrics, and read trends across separation from them. Absolute values of MCD and MVD are not comparable across studies that differ in either convention, and any replication should state both explicitly. Both metrics are also unnormalised sums over coordinates, so their absolute values grow with p even at constant per-coordinate distortion; comparisons across dimensionality should use per-coordinate or relative versions.

5.3 Future work

The central lesson generalises beyond the specific methods studied: synthetic data utility is task-relative, and a generator that preserves marginal resemblance need not preserve the higher-order geometry a clustering workflow depends on. Practically, the results argue for validating structural fidelity (distances, dispersion, separability, balance) against the intended analysis and against a resampling reference, and for treating a clustering comparison as a more demanding resemblance check than marginal summaries can provide. Natural extensions include other synthesisers, such as conditional inference trees (Hothorn, Hornik, and Zeileis, 2006), CART with larger minimum leaf sizes, parametric synthesisers, copulas and deep generative models; other clustering families, in particular Gaussian mixture models with the number of components chosen by BIC (Fraley and Raftery, 2002; Scrucca, Fop, Murphy, and Raftery, 2016), which can also return a single cluster, and non-convex and density-based methods; unequal group sizes and stronger correlations between features; varying N to probe small-sample behaviour; and a fuller study of the partition-level agreement introduced

in the case study, for example with transfer rules other than the nearest cluster mean. A promising direction is to model the discrepancy Δ of Eq. (1) directly as a learned function of the design factors, turning the utility surface itself into a predictive tool.

As a concrete next step, we propose applying the same comparison to the wearable dataset described in Vidiella Villegas and López (2025), to which we have direct access. Beyond providing a second and larger real-data test of the present design, this dataset would allow a comparison that neither study can perform alone. The present work evaluates synthesis as substitution, in which the synthetic dataset replaces the original. The applied study instead evaluates augmentation, in which the synthetic records are appended to the real ones and the union is clustered. These are different operations, and their effects are currently confounded with sample size: doubling the number of observations alters neighbourhood parameters and silhouette values independently of any distortion introduced by the generator. A three-way comparison of the partitions recovered from the real data, from the synthetic data alone, and from the augmented set, evaluated with the same metrics used here, would separate the effect of the generator from the effect of the additional sample. It would also make it possible to test whether the parameterisation of the generator, rather than the generator itself, accounts for the block artefact documented in this paper, since the applied study reports a substantial improvement in utility after replacing CART with conditional inference trees (Hothorn et al., 2006) and increasing the minimum leaf size. We regard this comparison as a natural continuation of the present study and as a practical way of establishing which of its conclusions generalise beyond the simulated setting.

6 Conclusions

The contribution of this study is empirical, but the comparison it uses is not tied to the simulation that illustrates it. The quantity $\Delta(\theta_R, \theta_S)$ is defined for any reference dataset and any generator, and the five factors varied here all have empirical counterparts: separation corresponds to how distinct the underlying groups are, the number of groups to the number of latent subpopulations, dimensionality to the number of recorded variables, correlation to the dependence between features, and the distribution family to the shape of the observed densities. In an applied dataset, dimensionality, feature correlations and marginal skewness can be computed directly, whereas the number of groups and their separation are properties of an unknown partition and can at best be estimated. Transferring the comparison to real data therefore requires two adaptations: the planted partition must be replaced by a reference obtained from the original data, and agreement must be calibrated against a resampling null, as done here. The diabetes case study (Section 4) makes both adaptations and reproduces the pattern of the simulation: agreement in the number of groups stays close to its resampling reference, while CART adds distortion to every geometric metric. Larger and more varied datasets would be informative rather than merely confirmatory. Real datasets may occupy regions of the design space that the present grid does not reach, for example correlations between related features well above the

maximum of $\rho = 0.4$ examined here, or stronger skewness than the Gamma family produces. They also exhibit properties the simulation excludes altogether, such as strongly unbalanced group sizes, ambiguous missingness in which a zero may indicate either genuine inactivity or a non-worn device, and heterogeneity between individuals. Each of these is a plausible source of additional distortion under sequential tree synthesis, and each would help to refine the conclusions reported here. We consider the unbalanced case particularly relevant, because the group-balance error ΔG and the hypothesis that the k-means criterion understates imbalance can only be fully tested when a substantial imbalance is present in the reference data.

Within the region of the design space studied here, synthetic data generated with CART can reproduce the appearance of clustering structure while subtly reshaping its geometry. Both algorithms recover well-separated structure reliably and struggle together when groups overlap. Where they diverge is in how often each reproduces, on synthetic data, the number of groups it would have found on the original data: k-means does so more often, on skewed data and not on Normal data, because its estimate is more stable under any resampling, not because CART damages it less. The utility of synthetic data for clustering is therefore best understood not as a property of the data but as a property of the data–generator–algorithm triple, to be verified against a resampling reference rather than assumed.

Supplementary information. The online supplementary material contains the recovery heatmap by number of groups and separation, the contingency table of original versus synthetic estimates, the sensitivity analysis for the candidate range of $\hat{k}$, the bootstrap null and the independent-pairs baseline by stratum, the geometric floor and scale-free metrics by stratum, the collapse cells, the clustering objectives, and the computational cost of the study.

Declarations

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Data availability: The simulated data can be regenerated exactly from the configuration file and seeds distributed with the code. The diabetes data of the case study are distributed with the R package `rrcov` (Todorov and Filzmoser, 2009).

Code availability: The full pipeline, from data generation through to the figures and tables reported here, is available at https://github.com/Sterrysx/synthetic-clustering. Every stage is driven from a single configuration file and can be re-executed end to end.

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